# Calculus and Number Systems from Differential Propagation:

## *A Mechanism-First Derivation*

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**Abstract**

Every standard treatment of calculus and the number systems begins after an invisible decision: the real number line, the limit concept, and the axioms of arithmetic are imported as foundations before derivation begins. This paper demonstrates that all of these can instead be derived from a single dynamical mechanism — differential propagation — without import.

The mechanism: relational patternings propagate through gradient fields, accumulating loaded history that sets their gradient demands. Simpler patterns propagate faster. When demands exceed available gradients, patterns reconfigure toward coherence. From this, with no additional axioms: the natural numbers emerge as counts of propagation steps; integers as bidirectional reconfiguration; rationals as load ratios; reals as coherence completion; complex numbers as the minimum carrier extension satisfying $G^2=G_{neg}$; the differential ε-nilpotency as the boundary condition of first-order propagation; Taylor coefficients as symmetry normalisations; integration as coherence accumulation; the Fundamental Theorem of Calculus as $G_{diff} \circ G_{int} = G_{id}$.

sin and cos are not primitive — they are the fixed points of $G_{diff}{}^4$, the 4-step orbit of the rotation gradient in $\mathbb{C}$. The constants e, π, φ are each a different kind of fixed point of their respective gradient families. The Gaussian integral $\int e^{-x^2}dx = \sqrt{\pi}$ is the structural collision of $G_{diff}$'s fixed point with $G_{rot}$'s coherence cycle. Shannon entropy is the propagation cost measure in the probability carrier. Period-doubling cascades to chaos are Observation 2.2 made dynamically visible.

In the computational carrier, the Halting Problem is Observation 2.2 instantiated over Turing machines — the same structure as Gödel incompleteness, with the Liar a distinct case involving value rather than history self-reference. The mechanism distinguishes these and predicts load growth rate $2^d$ per simulation depth. In the algebraic carrier, the quintic's unsolvability follows from $A_5$ generating irreducible non-commutative drag. Mathematical truth is coherence: a theorem is a pattern that propagation cannot destroy.

A Python implementation (github.com/[repository]) verifies all structural claims to machine precision.

One point requires explicit statement: gradient fields G are not a different kind of thing from patterns P. G is a pattern occupying the gradient role contextually — a local assignment, not an intrinsic substance difference. G_diff, G_rot, G_×, and every other gradient family introduced in this paper are patterns in the space of gradient families, propagating through a meta-context when studied. This G/P distinction, developed fully in [Pugmire, 2026a] Section 2.6, closes the

ontological loop of the mechanism and prevents gradient fields from being smuggled in as hidden substances.

**Keywords:** differential propagation, foundations of calculus, number systems, dual numbers, Taylor series, fundamental theorem of calculus, halting problem, linear logic, relevance logic, automatic differentiation, Galois theory, coherence, fixed points

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### Note on the Subject-Predicate Cut

Standard mathematics begins after a cut: the decision to treat mathematical objects as substances with intrinsic properties, prior to and independent of the operations that generate them. The history of mathematics follows a uniform pattern: framework establishes identity through invariance, operations generate boundary-violating objects (irrationals, imaginaries, infinitesimals), practitioners provisionally treat them as existing, recover coherent results, then reconstruct the framework. Operations are more fundamental than objects. Identity emerges from operational closure, not prior substance. This paper follows that insight to its conclusion.

The same reification that applies to patterns applies equally to gradient fields. The standard notation $P/G \to Q$ presents G as if it occupies a different ontological position from P. It does not. A gradient field G is a pattern that has been contextually assigned the gradient role in a specific interaction. G has loaded history. G was constituted by prior propagation. G can become P in a higher-order propagation event. When this paper defines $G_{diff}$, $G_{rot}$, $G_{\times}$ and derives results from them, it is treating these gradient families as objects of study — patterns propagating through a meta-context. The description of the mechanism is itself an instance of the mechanism. There is no view from outside.

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## 1. Introduction

Standard foundations proceed axiomatically: Peano axioms, field axioms, completeness axiom, then calculus via ε-δ limits. This paper does not dispute the correctness of the resulting mathematics. It disputes the claim that axioms are the deepest starting point. Every axiom in the standard development is the boundary condition of a specific propagation regime.

The mechanism from which everything is derived is differential propagation [Pugmire, 2026a]. As relational patternings propagate through gradient fields, they accumulate loaded history that sets their gradient demands. Patterns with lower loaded history propagate faster. When demands exceed available gradients, patterns reconfigure toward coherence.

### 1.1 Four Types of Claim

The paper makes four distinct types of claim with different epistemic weights.

| Claim Types |
| --- |
| **A. Derives:** The mechanism independently generates the structure. Natural numbers, integers, rationals, linear logic consumption state, relevance logic gradient connectivity, flux propagation. No external import. |

**B. Motivates and constructs:** The mechanism explains why the structure is needed and provides the natural construction. The reals: the mechanism derives the pressure; Cauchy construction follows.

**C. Reveals structure:** The mechanism reframes a known result to make its structural role visible. FTC as $G_{diff} \circ G_{int} = G_{id}$ is the same structure as double negation in logic. The analogy is the contribution.

**D. Asks the right question:** Complex numbers from 'what gradient field G satisfies $G \circ G = G_{neg}$?'. The question forces the carrier extension.

### 1.2 Note on Mutual Grounding

**Section 2 uses scalar magnitudes for expository clarity**

The mechanism's conceptual primitive is qualitative: one propagation step, an ordering on accumulated histories, a coherence threshold. The scalar formulas ($L_P$, rate, demand) are the natural quantification of this structure using $\mathbb{N}$ and $\mathbb{Q}$, derived in Section 3. Neither section is logically prior; they co-constitute each other. This is mutual grounding, not circular reasoning.

## 2. The Mechanism of Differential Propagation

### 2.1 Loaded Patterns and Contexts

**Definition 2.1 — Loaded Pattern**

$P = (v_P, L_P)$ where $v_P \in V$ is the designation component and $L_P = |H_P| \geq 0$ is the informational load. $L_P = 0$: seed state. The components $v_P$ and $L_P$ must not be conflated.

**Definition 2.2 — Context**

$C = (\Gamma_C, o_C, \theta_C)$. Support: $\min(L_P, \theta_C)$. Demand: $\max(0, L_P - \theta_C)$. Coherent iff demand = 0. Valid iff designated and coherent.

**Definition 2.3 — Propagation Operator**

$P /_C G := Q$. G specifies independently: (a) value transformation $v_P \to v_Q$, and (b) load rule $L_P \to L_Q$. For binary operations, $L_Q = o_C(L_P, L_Q)$. For unary operations G defines $L_Q$ directly. The load rule is a property of the gradient field, not the operator.

**Note on the G/P Distinction**

The notation $P/G \to Q$ can suggest that G is a different kind of thing from P. It is not. G is a pattern occupying the gradient role contextually — a local and momentary assignment at the moment of interaction.

**All G is P post-boundary-imposition.** Every gradient family in this paper — $G_{diff}$, $G_{rot}$, $G_{\times}$,

$G_{int}$, $G_{neg}$ — is a pattern in the space of gradient families. These are objects of study here, propagating through a meta-context when defined and examined. $G_{diff}$ has loaded history. $G_{rot}$ was constituted by prior propagation. Each can become P in a higher-order event.

**The P/G distinction is local and momentary.** When P and G meet in context C, one contextually occupies the gradient role. After the interaction, roles can flip. What was G becomes P for the next propagation event. The distinction is a relational cut at the moment of interaction, not a permanent substance difference.

**Meta-claims collapse back.** 'The mechanism generates ℝ from ℚ' is itself P/G→Q running in a meta-context whose carrier is the space of number systems. The description of the mechanism is an instance of the mechanism. There is no view from outside. The full treatment is in [Pugmire, 2026a] Section 2.6.

**Theorem 2.1 — Propagation Ordering**

Among incoherent patterns: rate(P,C) > rate(Q,C) ⟺ $L_P < L_Q$. Simpler patterns propagate faster. Proof: rate = $\theta_C/L$ for incoherent patterns. □

**Observation 2.2 — Maximum Complexity**

When $H_P \supseteq H_P$: $L \to L + L \to \infty$. No finite $\theta_C$ contains this demand. Self-referential loaded history generates unbounded complexity. Simpler patterns propagate outward from the complex core.

### 2.2 Drag Regimes as State Constraints

The nonzero-drag regimes encode boundary conditions of their gradient families as state constraints on patterns, not continuous load formulas. This is the key correction over earlier formulations using $|L_P - L_Q|$ terms, which incorrectly give zero penalty to equal-load disconnected patterns.

**Definition 2.4 — Drag Regimes**

**Zero-drag (classical, calculus):** $o_C(L_P, L_Q) = L_P + L_Q$.

**Linear drag:** Each pattern carries available ∈ {True, False}. First co-propagation marks it consumed. Second use raises LinearLogicViolation (demand = ∞). The state change IS the constraint.

**Relevance drag:** Patterns carry history_tags. Co-propagation requires history(P) ∩ history(Q) ≠ ∅. Disconnected patterns raise RelevanceLogicViolation (demand = ∞). Variable sharing = gradient connectivity.

## 3. Number Systems as Propagation Structures

### 3.1 Natural Numbers (Type A)

**Definition 3.1 — Natural Numbers**

ℕ is generated from the seed state by the propagation step:

```
0 := seed (Lₚ = 0)
succ(n) := n ⊕ 1  (one additional propagation step)
m + n := m steps then n steps (loads accumulate)
m × n := m-step propagation repeated n times
```

The successor relation IS a propagation step. Zero IS the empty loaded history. Peano axioms are boundary conditions of this construction.

## 3.2 Integers (Type A)

ℤ extends ℕ with reconfiguration direction. Negative integers are signed reconfiguration directions, not new constructions. n + (−n) = 0: reconfiguration undone. The product of two negative reconfigurations is positive — forced by reconfiguration composition.

## 3.3 Rationals and the Stern-Brocot Tree (Type A)

The propagation rate is already a ratio: rate(P, C) = min($L_P$, θ)/$L_P$. Rational numbers are exactly the ratios arising as propagation rates.

**Theorem 3.2 — Density.** Between any two distinct rationals, their mediant lies strictly between them: (a+c)/(b+d) satisfies a/b < (a+c)/(b+d) < c/d. Proof: the mediant has the minimum possible denominator for a fraction strictly between a/b and c/d. It is the minimum-load fraction in that interval. □

The Stern-Brocot tree is Theorem 2.1 drawn as a binary picture. Every positive rational appears exactly once, at depth equal to its denominator complexity (propagation cost). The tree is constructed by mediants: each new fraction is the minimum-load fraction between its parent fractions. At depth d, fractions have denominator ≤ d+1. Continued fractions are the navigation paths to irrationals: each term is the number of steps in one direction before turning.

**Example.** e = [2; 1, 2, 1, 1, 4, 1, 1, 6, …]: every third term is 2k (k=1,2,…). This regular pattern reflects e's special status as the $G_{diff}$ fixed point — e has a regular loaded history accumulation in ℚ, unlike π which has irregular terms [3; 7, 15, 1, 292, …].

## 3.4 Real Numbers (Type B)

**Definition 3.4 — Reals as Coherence Completion**

The mechanism contributes: (i) the need — Newton reconfiguration for $x^2=2$ generates a Cauchy sequence in ℚ converging to no rational (Theorem 3.3); (ii) the natural construction — x ∈ ℝ iff there exists $\{q_n\} \subset \mathbb{Q}$ with demand($P_n$, $C_\theta$) → 0 for all θ > 0. The resulting structure is standard Cauchy completion.

## 3.5 Complex Numbers (Type D)

**Definition 3.5 — Complex Numbers from $G^2 = G_{neg}$**

The mechanism asks: what gradient field G satisfies $G \circ G = G_{neg}$? Over $\mathbb{R}$: no such G exists ($c^2 \geq 0$ for all $c \in \mathbb{R}$). This generates reconfiguration pressure. The minimum carrier extension admitting a solution adds direction i with $i^2 = -1$. The rotation gradient $G_{rot}(a,b) = (-b, a)$ satisfies $G_{rot}(G_{rot}(a,b)) = (-a,-b) = G_{neg}(a,b)$. Verified: i is not defined; it is the minimum carrier extension satisfying the demand. □

### 3.5.1 sin and cos as $G_diff^4$ Fixed Points

Play sessions revealed: sin and cos are not primitive. They emerge as the real projections of the 4-step orbit of the rotation gradient in $\mathbb{C}$.

**Theorem 3.4 — Trigonometric Functions as Fixed Points**

The equation $f^{(n)} = f$ (f is its own nth derivative) has solutions $\exp(\lambda x)$ where $\lambda^n = 1$.

n=1: $\lambda=1 \rightarrow \exp(x)$ [$G_{diff}$ fixed point in $\mathbb{R}$]

n=2: $\lambda=\pm1 \rightarrow \cosh(x), \sinh(x)$ [$G_{diff}^2$ fixed points]

n=4: $\lambda=\pm1,\pm i \rightarrow \cos(x), \sin(x), \exp(\pm x)$ [$G_{diff}^4$ fixed points in $\mathbb{C}$ projected to $\mathbb{R}$]

sin and cos are the REAL PARTS of the 4-step rotation orbit. They are not primitive — they emerge when the rotation carrier extension ($\mathbb{C}$) is projected onto $\mathbb{R}$. The carrier hierarchy determines the fixed-point hierarchy. □

**Theorem 3.5 — Euler's Formula**

$\exp(i\theta) = \cos(\theta) + i\cdot\sin(\theta)$. Proof: exp on $\mathbb{C}$ is defined by the Taylor series $\Sigma\, z^n/n!$. Differentiating $f(\theta) = \exp(i\theta)$: $f' = i\cdot f$, $f(0)=1$. Decomposing $R+iI$: $R'=-I$, $I'=R$ with $R(0)=1$, $I(0)=0$. Unique solution: $R=\cos$, $I=\sin$. At $\theta=\pi$: $\exp(i\pi)+1=0$. No circular assumption; follows from Taylor series (Section 5) and the rotation gradient's fixed-point property. □

## 3.6 The Three Coherence Constants

**e, π, φ: Three Different Kinds of Fixed Points**

**e = fixed point of $G_{diff}$ in $\mathbb{R}$:** $d/dx[e^x] = e^x$. The exp function is its own derivative. e is also the most efficient propagation base ($e^{1/e}$ is maximal over all $b^{1/b}$), the natural unit of information (Shannon entropy uses log base e), and the coefficient sum of all symmetric propagation orbits: $\Sigma\, 1/n! = e$.

**π = coherence cycle of $G_{rot}$ in $\mathbb{C}$:** one full rotation in $\mathbb{C}$ costs $2\pi$. π appears wherever the rotation gradient generates a limit: circle area, Gaussian integral, Basel sum $\Sigma\, 1/n^2 = \pi^2/6$, Catalan asymptotics $C_n \sim 4^n/(n^{3/2}\sqrt{\pi})$.

**φ = fixed point of ratio reconfiguration in $\mathbb{Q}$:** $x \rightarrow 1+1/x$ has fixed point $\varphi = (1+\sqrt{5})/2$. Fibonacci numbers are the propagation history of this reconfiguration; φ is the coherence limit. φ appears wherever self-similar ratio growth generates a fixed point.

*These are not three aspects of the same constant. They are three structurally different fixed-point types: $G_{diff}$ fixed point (e); orbit coherence constant (π); ratio reconfiguration fixed point (φ). The mechanism distinguishes them precisely where numerology groups them.*

### 3.7 Primes as Seeds of the Multiplicative Carrier (Type A/C)

The multiplicative carrier is generated by $G_\times$: (a,b) → a·b. A prime is a SEED: not reachable by $G_\times$ from smaller patterns.

**Fundamental Theorem of Arithmetic (Type A)**

Every positive integer n ≥ 2 has a unique prime factorization. Every loaded history in the multiplicative carrier decomposes uniquely into seed-state contributions. The prime factorization IS the loaded history; $\Sigma\, k_i \cdot \ln(p_i) = \ln(n)$ exactly for all n. Verified: ln(12) = 2·ln(2)+ln(3) = 2.48490… exactly; holds for all tested n.

**Prime Number Theorem (Type C — Reveals Structure)**

**What the mechanism contributes:** (i) ln(n) IS the propagation cost in the multiplicative carrier — exact, not approximate. (ii) The Euler product $\zeta(s) = \Pi_p\, 1/(1-p^{-s})$ is the partition function of the multiplicative carrier; FTA is the product formula. (iii) The PNT is equivalent to ζ having no zeros on Re(s)=1 — structural insight: prime distribution is determined by the coherence structure of ζ in the complex carrier.

**What the mechanism does not derive:** The proof that ζ has no zeros on Re(s)=1 requires complex analysis (Hadamard, de la Vallée Poussin). The mechanism motivates why primes thin out and identifies the structural connection (ζ = partition function), but the 1/ln(x) rate is standard analysis. Claim type: C.

## 4. The ε-Structure of Differentiation

### 4.1 Nilpotency as Boundary Condition

**Theorem 4.1 — ε-Nilpotency**

The first-order propagation regime is defined as the regime where second-order terms are negligible relative to first-order terms. Within this regime, $\varepsilon^2=0$ is the DEFINING condition, not a derived consequence. Demonstration: at step size δ, $f(x+\delta)-f(x) = f'(x)\delta + f''(x)\delta^2/2$. Ratio $\varepsilon^2/\varepsilon = \delta \to 0$. The first-order regime is self-consistent: treating $\varepsilon^2=0$ introduces error of order $\varepsilon^2$ relative to terms of order ε.

### 4.2 Dual Numbers from Loaded Patterns

**Definition 4.1 — Dual Number**

P = (v, e) where v is the value component and e is the first-order loaded history. Arithmetic

rules are gradient field definitions derived from zero-drag co-propagation:

```
Addition:  (v₁+v₂, e₁+e₂)          [histories add independently]
Product:   (v₁v₂, e₁v₂+v₁e₂)      [Leibniz; e₁·e₂=0 boundary condition]
exp:       (e^v, e·e^v)              [val ≡ ε-component: the G_diff fixed point]
sin:       (sin(v), e·cos(v))        [chain rule: sin and cos are G_diff^4 orbits]
```

The exp fixed point: val = e-component at every point. This is what $G_{diff}$ fixed point means in the dual number representation.

### 4.3 Verification

| Expression | At x = | Mechanism ε | Analytical | Error |
|---|---|---|---|---|
| $d/dx[x^3]$ | 1.3 | 5.07000000 | 5.07000000 | 0.00e+00 |
| $d/dx[x^2\sin(x)]$ | 1.3 | 2.95732430 | 2.95732430 | 0.00e+00 |
| $d/dx[\sin(x^2)]$ | 1.3 | −0.30919608 | −0.30919608 | 0.00e+00 |
| $d/dx[e^x/x]$ | 1.3 | 0.65135444 | 0.65135444 | 2.22e−16 |
| exp fixed point | 1.0 | val = e = L | val = e = L | 0.00e+00 |

*Table 4.1: Gradient field rules verified. All errors zero or at machine precision.*

## 5. Taylor Coefficients and the Fourier Connection

### 5.1 1/n! as Symmetry Normalisation

**Theorem 5.1 — Taylor Coefficients**

In the zero-drag regime, gradient fields commute: $\varepsilon_1 \cdot \varepsilon_2 = \varepsilon_2 \cdot \varepsilon_1$. Given commutativity, n propagation steps in direction h produce n! equivalent orderings, each giving the same loaded contribution $f^{(n)}(x) \cdot h^n$. The normalisation 1/n! counts each configuration once. The commutativity is derived from zero-drag; the counting of n! orderings uses $S_n$ (imported). If drag ≠ 0: steps do not commute; 1/n! is replaced by $1/[n]_q!$ (see Section 7.3). □

### 5.2 Fourier Series as Taylor in the Rotation Regime

Play sessions revealed that Taylor and Fourier are the same expansion mechanism in different gradient families.

**Observation 5.1 — Taylor and Fourier as the Same Operation**

**Taylor:** expansion in the ε-direction (linear gradient $G_\varepsilon$). Basis: fixed points of $G_\varepsilon = \{x^n/n!\}$. Coefficients: nth loaded history of f at x.

**Fourier:** expansion in the iθ-direction (rotation gradient $G_{rot}$). Basis: fixed points of $G_{rot} = \{e^{inx}\}$. Coefficients: nth loaded history of f along the rotation orbit.

The difference is the gradient family, not the operation. Both expand a function's loaded history in basis functions that are fixed points of their respective gradient families. The Fourier basis $\{e^{inx}\}$ IS the $G_{rot}$ orbit — the rotation gradient's fixed points, verified: $e^{i\theta} = \cos(\theta) + i \cdot \sin(\theta)$ (Theorem 3.5).

**Gibbs phenomenon:** at a discontinuity, the loaded history generates high-frequency components that no finite frequency count can contain. Coherence failure at a discontinuity is structural, not numerical — it has the same cause as the Feigenbaum overshoot (Section 10.1): a structural property of coherence-seeking mechanisms at boundaries.

# 6. Integration as Coherence Accumulation

## 6.1 Definition Without Limits

**Definition 6.1 — Integration as Coherence Accumulation**

$\int_a^b f\,dx$ is the unique value S such that, for any coherence threshold $\theta > 0$, partition refinements reach $|S_n - S_{n-1}| < \theta$ for all sufficiently fine partitions. Each partition step is a propagation event. No completed infinity. No substance called a limit.

## 6.2 The Fundamental Theorem as Structural Identity

**Theorem 6.1 — FTC (Type C — Reveals Structure)**

$G_{diff} \circ G_{int} = G_{id}$.

The FTC has a standard proof. The mechanism reveals its structural form: opposite gradient operators composing to identity. This is the same structure as double negation elimination $G_{neg} \circ G_{neg} = G_{id}$ in logic — both instances of a structural property of the propagation operator. The analogy is the contribution; the proof is standard.

## 6.3 The Gaussian Integral: e Meets π at the Quadratic Gradient

The result $\int e^{-x^2}dx = \sqrt{\pi}$ appears mysterious because e and π seem unrelated. The mechanism shows they are connected through the quadratic gradient field $G_{quad}$: $x \to x^2$.

**Theorem 6.2 — Gaussian Integral**

Let $I = \int_{-\infty}^{\infty} e^{-x^2}\,dx$. Co-propagate two independent copies (zero-drag):

$$I^2 = (\int e^{-x^2}dx)(\int e^{-y^2}dy) = \iint e^{-(x^2+y^2)}\,dxdy$$

Convert using $G_{rot}$ (the rotation gradient's area element $dxdy = r \cdot dr \cdot d\theta$):

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2}\,r\,dr\,d\theta = 2\pi \times (1/2) = \pi$$

The 2π factor IS $G_{rot}$'s coherence cycle (Section 3.6). The 1/2 is $\int_0^{\infty} e^{-r^2}r\,dr = 1/2$ by substitution. Therefore $I = \sqrt{\pi}$.

$G_{diff}$ fixed point (e) × $G_{rot}$ coherence cycle (π) = coherence in $\mathbb{R}^2$. The normal distribution $N(0,1) = (1/\sqrt{(2\pi)}) \cdot e^{-x^2/2}$ is forced by this collision. It is the maximum entropy distribution on $\mathbb{R}$ with fixed variance — the coherence state of $\mathbb{R}^2$ under the combined gradient. □

## 6.4 Euler's Basel Proof: Seeds of the Rotation Carrier

Euler's proof that $\Sigma\, 1/n^2 = \pi^2/6$ connects the integer zeros of sin (the seeds of $G_{rot}$) to π.

**Theorem 6.3 — Basel Problem via Rotation Seeds**

$\sin(\pi x)$ has zeros at all integers: the coherence points of $G_{rot}$ in $\mathbb{R}$. Euler's product formula: $\sin(\pi x) = \pi x \times \Pi_{n\geq1}(1 - x^2/n^2)$. Expanding the product, the coefficient of $x^2$ is $-\Sigma\ 1/n^2$. The Taylor series gives $\sin(\pi x)/(\pi x) = 1 - \pi^2x^2/6 + \ldots$, so the $x^2$ coefficient is $-\pi^2/6$. Matching: $\Sigma\ 1/n^2 = \pi^2/6$.

Seeds determine constant. Constant encodes seeds. $\pi^2/6$ is the total inverse-squared distance to the rotation gradient's zero structure. Verified: $\Sigma_{n=1}^{10000}\ 1/n^2 = 1.64483\ldots$, $\pi^2/6 = 1.64493\ldots$, error $10^{-4}$. □

## 6.5 Euler-Lagrange as Reconfiguration in Function Space

Definition 2.6 (reconfiguration) extends to the infinite-dimensional function-space carrier.

**Theorem 6.4 — Euler-Lagrange as Definition 2.6**

Carrier: smooth curves $y: [0,1] \to \mathbb{R}$. Gradient demand: the action $S = \int L(y, y', x)\, dx$. Reconfiguration step: $y \to y + \delta y$ where $\delta y$ minimises demand.

The fixed-point condition $\delta S = 0$ gives the Euler-Lagrange equation: $\partial L/\partial y - d/dx[\partial L/\partial y'] = 0$.

The brachistochrone: minimum slide time from (0,0) to (1,−1). The cycloid achieves minimum action. Verified: cycloid time < straight line time.

**Demonstration — Schrödinger as Definition 2.6**

**Carrier:** $V = L^2(\mathbb{R}, \mathbb{C})$ — square-integrable complex wavefunctions $\psi$.

**Gradient demand:** action $S = \int\langle\psi|\hat{H}|\psi\rangle\, dt$. Reconfiguration: $\psi \to \psi + \delta\psi$ minimising demand.

**Fixed-point condition $\delta S = 0$:** using Lagrangian $L = \langle\psi|(i\hbar\partial/\partial t - \hat{H})|\psi\rangle$ and varying over $\psi^*$:

```
iħ∂ψ/∂t = Ĥψ   (Schrödinger equation)
```

This IS Definition 2.6: $\psi^* = \text{argmin}\{\text{demand}(\psi', C)\}$ where demand = deviation from minimum action. The Schrödinger equation is the reconfiguration condition in the $L^2$ carrier.

**The same structure:** Maxwell (carrier = gauge fields $A_\mu$; $\delta S=0 \to \partial_\nu F^{\mu\nu} = J^\mu$); Einstein (carrier = metrics $g_{\mu\nu}$; $\delta S=0 \to G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\, T_{\mu\nu}$). Every physical law in Lagrangian form is Definition 2.6. Conservation laws are gradient fields leaving S invariant (Section 10.5).

# 7. Nonzero-Drag Regimes, q-Calculus, and Galois Theory

## 7.1 Linear Logic: Consumption State (Type A)

**Definition 7.1 — Consumption State**

Each pattern carries available ∈ {True, False}. First co-propagation marks it consumed: available → False. Attempting to co-propagate a consumed pattern raises LinearLogicViolation (demand = ∞). The !A operator marks a pattern as infinitely copyable

(zero-drag applied to that specific pattern).

**Verified:** consumed pattern raises LinearLogicViolation on second use. Available patterns combine additively. Correction over formula-based approach: $|L_P - L_Q|$ gives zero penalty to equal-load disconnected patterns; the state constraint correctly generates infinite demand.

## 7.2 Relevance Logic: Gradient Connectivity (Type A)

**Definition 7.2 — Gradient Connectivity**

Each pattern carries history_tags: the gradient operations applied to it. Two patterns are connected iff history(P) ∩ history(Q) ≠ ∅. Co-propagation of disconnected patterns raises RelevanceLogicViolation (demand = ∞). Variable sharing IS gradient connectivity: shared variable means both patterns propagated through that gradient (Erlangen insight: shared gradient ancestry = shared identity frame).

**Verified:** patterns with history_tags {'x'} and {'y'} raise RelevanceLogicViolation. Shared tag 'x' allows additive combination.

## 7.3 q-Calculus as Partial Drag Regime

When drag ≠ 0, propagation steps partially commute. This deforms the counting structure of the Taylor coefficients.

**Definition 7.3 — q-Derivative and q-Numbers**

The q-derivative: $D_q f(x) = (f(qx) - f(x))/((q-1)x)$. At q=1: $D_q f(x) \to f'(x)$ (L'Hôpital). At q≠1: a deformed derivative satisfying the q-Leibniz rule:

```
D_q(fg)(x) = f(qx)·D_q(g)(x) + g(x)·D_q(f)(x)
```

This is NOT symmetric in f and g (unless q=1) — the partial drag shows: left application ≠ right application. The q-numbers $[n]_q = (q^n-1)/(q-1) = 1+q+\ldots+q^{n-1}$ are counting in the partial-drag regime. At q=1: $[n]_q = n$ exactly. q-factorial $[n]_q! = [1]_q \cdot [2]_q \cdot \ldots \cdot [n]_q$ replaces n! in the q-Taylor series. Verified: q-power rule $D_q[x^n] = [n]_q \cdot x^{n-1}$ holds exactly.

## 7.4 Galois Theory: Non-Commutative Drag in the Algebraic Carrier

The connection between drag regimes and algebraic solvability: solving by radicals requires a chain of commuting gradient applications. When the symmetry group has irreducible non-commutativity, no such chain exists.

**Theorem 7.1 — Quintic Unsolvability as Irreducible Drag**

Solving by radicals = reaching coherence through a chain of gradient fields, each commuting with the symmetry at that stage (zero drag at each level). The Galois group of a polynomial is the symmetry group of its roots.

For degree ≤ 4: the Galois group is a subgroup of $S_4$. $S_4$ is solvable: $S_4 \supset A_4 \supset V_4 \supset \{e\}$,

each quotient abelian (commutative). Zero drag at each level → solvable by radicals.

For generic degree 5: Galois group = $S_5$. $S_5 \supset A_5 \supset \{e\}$. $A_5$ is simple: no normal subgroups; non-abelian. The solvability chain breaks at $A_5$. $A_5$ generates IRREDUCIBLE non-commutative drag — no sequence of commuting radical operations can reduce it. The quintic is not solvable by radicals because the algebraic carrier has non-trivial drag that radical extensions cannot reduce.

$|S_2|=2$, $|S_3|=6$, $|S_4|=24$, $|S_5|=120$ verified computationally. The gap between solvable (degree ≤ 4) and non-solvable (degree 5) is the gap between zero-drag and non-zero-drag gradient chains. □

## 8. Flux Propagation Through Iterative Solvers

In the calculus regime, the loaded history IS the flux — the propagating gradient. The flux field is not metadata; it is a second propagating quantity governed by the same arithmetic as val [Pugmire, 2026b].

**Definition 8.1 — Solve-to-Coherence**

Seed the input parameter a with unit flux: Pattern(a, 1.0). Seed the solver state with zero flux: Pattern(init, 0.0). Iterate until $|val_{k+1} - val_k| < \theta$. At convergence, flux carries the exact gradient.

The gradient arrives WITH the solution — it propagated throughout. Memory: 48 bytes (two floats), constant regardless of iteration depth. No computation graph. No Jacobian inversion.

**Flux convergence:** at fixed point $x^* = f(x^*, a)$ with $|\partial f/\partial x| < 1$, the flux $g^* = (\partial f/\partial a)/(1 - \partial f/\partial x)$ — the IFT gradient, reached iteratively without matrix inversion [Pugmire, 2026b, Appendix A].

| Solver | a | val | Flux (mechanism) | Exact gradient | Error | Iters |
|---|---|---|---|---|---|---|
| Babylonian √a | 4 | 2.000000 | 0.250000 | 1/(2√4) | 0.00e+00 | 6 |
| Newton ∛a | 8 | 2.000000 | 0.083333 | (1/3)·8^{-2/3} | 1.39e−17 | 8 |
| Implicit F(x,a) | 0 | 1.000000 | −0.333333 | −x*/(3x*²+a) | 0.00e+00 | 3 |
| Cosine FP | — | 0.739085 | −0.402490 | −sin(x*)/(1+sin(x*)) | 1.82e−11 | 69 |
| Stiff ODE k=1000 | k | 0.009901 | −9.80e−6 | −dt·y/(1+k·dt)² | 0.00e+00 | 4 |

*Table 8.1: Flux propagation benchmarks. All agree with analytical gradients to machine precision. Memory constant at 48 bytes regardless of iteration depth.*

## 9. The Halting Problem and Observation 2.2

The Halting Problem is Observation 2.2 instantiated in the computational carrier. The mechanism does not independently prove undecidability; Turing's diagonal argument proves

that. The mechanism provides the structural account of what that proof demonstrates, and makes three specific verifiable predictions.

## 9.1 Programs as Loaded Patterns

**Definition 9.1 — Computational Carrier**

P = (state, load): state is the current computation state; load is accumulated execution steps. Execution: $P / G_{step} \rightarrow Q$. Halting: fixed point. Non-halting: load $\rightarrow \infty$. Terminating programs (countdown, Fibonacci) have load converging to a finite value — halting detectable via coherence.

## 9.2 H(H) Instantiates Observation 2.2

**Theorem 9.1 — Self-Referential Load Divergence**

A halting decider $G_H$ must contain P's full execution history. Applied to H: $H_{G_h(H)} \supseteq H_{G_h(H)}$. Load at simulation depth d: $L_d = 2^d \rightarrow \infty$. No finite $\theta_C$ contains this. $G_H$ cannot complete on input H.

Turing's diagonal constructs D: if H(D) says D halts, D runs forever; if not, D halts. In mechanism terms: $H_D \supseteq H_{H(D)} \supseteq H_D$. The contradiction is the symptom; self-referential loaded history is the cause. □

**Grounding the $2^d$ rate in UTM resource usage:** For a complete halting decider H of description length n, simulation at depth d requires representing ALL computational paths H must consider at that depth. Because H decides halting for all programs of length ≤ n, and each execution step branches binary (each memory bit is 0 or 1), at depth d there are at least $2^d$ distinct computation paths requiring representation. Tape usage $L_d \geq 2^d \cdot n$. The abstract L = L+L model is not stipulated; it follows from what a complete oracle must physically represent. Simple non-self-referential programs (countdown, Fibonacci) have finite simulation trees and remain decidable.

## 9.3 The Liar Paradox: A Different Structure

**Theorem 9.2 — Value vs History Self-Reference**

Pattern P = (v, L) has two components. Self-reference can hit either:

```
Liar (L = ¬L):  v = 1-v  ← value self-reference: no fixed point in {0,1}
                 L = L   ← load trivially satisfied: no load divergence
Gödel/Turing:   H_G ⊇ H_G  ← history self-reference: L → 2^d → ∞
```

Gödel and Turing are the same structure (Observation 2.2 in different carriers). The Liar is distinct: value impossibility, no load divergence. There is no route from L = ¬L to load divergence without adding a process structure not in the definition. This is a stronger result than claiming all three are identical.

| Case | Carrier | Component | Form | Consequence | Load diverges? |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| Liar | {0,1} | Value (v) | v = 1−v | No fixed point in V | No |
| Gödel | PA+Prov | History (L) | H_G ⊇ H_G | True, unprovable | Yes: $2^d$ |
| Turing | TM | History (L) | H_H ⊇ H_H | H cannot decide itself | Yes: $2^d$ |

*Table 9.1: Taxonomy of self-referential impossibilities. Gödel and Turing are the same structure; the Liar is distinct.*

## 9.4 What the Mechanism Adds Over the Diagonal Argument

**Additions Over the Diagonal**

**(a) Rate prediction:** Load at depth d is $L_d = 2^d$. The diagonal shows failure exists; the mechanism predicts the exponential rate. Verified computationally.

**(b) Structural boundary:** The diagonal constructs an adversarial input for any H. The mechanism characterizes the boundary: $H_P \supseteq H_P$ is the condition identifying which programs generate undecidability.

**(c) Taxonomy:** The diagonal technique applies to both the Liar and Gödel/Turing without distinguishing the type of self-reference. The mechanism distinguishes them structurally.

**Open question:** Whether (a) and (b) constitute a deeper explanation than the diagonal, or a complementary characterisation, is a philosophical question stated explicitly rather than assumed.

## 9.5 Cantor's Diagonal and p-Adic Numbers

The mechanism unifies several results.

**Observation 9.1 — Cantor's Uncountability as Type D Carrier Extension**

Cantor's diagonal is not Observation 2.2 (iterative load doubling). It is a one-shot construction, not an iterating process. The correct framing: it is the same carrier-extension pressure as $\mathbb{N} \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R} \to \mathbb{C}$.

The mechanism asks: what carrier contains ALL subsets of $\mathbb{N}$? Cantor's diagonal shows no map $f: \mathbb{N} \to P(\mathbb{N})$ is surjective: construct $d = \{n \in \mathbb{N} : n \notin f(n)\}$. Then $d \notin \text{image}(f)$. The power set operation generates an entity (d) outside any proposed listing — the same carrier-extension pressure as subtraction generating negatives outside $\mathbb{N}$, or limits generating irrationals outside $\mathbb{Q}$.

This is TYPE D: asking 'what carrier contains all subsets of $\mathbb{N}$?' forces the extension to $P(\mathbb{N})$ as the minimum carrier satisfying the demand. The hierarchy $\aleph_0 < \aleph_1 < \ldots$ is successive carrier extensions under diagonalisation pressure, exactly as $\mathbb{N} < \mathbb{Z} < \mathbb{Q} < \mathbb{R} < \mathbb{C}$ is successive carrier extensions under operational pressure.

**Observation 9.2 — p-Adic Numbers as Prime-Centric Coherence**

The p-adic absolute value $|n|_p = p^{-v_p(n)}$ defines a prime-centric drag regime: divisibility by p = low gradient demand. The coherence completion under this regime gives $\mathbb{Q}_p$.

Ostrowski's theorem: every non-trivial absolute value on $\mathbb{Q}$ is either the standard Archimedean value (giving $\mathbb{R}$) or the p-adic value for some prime p (giving $\mathbb{Q}_p$). There are exactly (primes + 1) distinct drag regimes on $\mathbb{Q}$.

Product formula: $\Pi_p |n|_p \cdot |n|_\infty = 1$ for all $n \in \mathbb{Q}\times$. Total loaded history across all coherence regimes is balanced. Verified: n=12: $|12|_\infty \times |12|_2 \times |12|_3 = 12 \times 0.25 \times 0.333 = 1.000$ exactly.

**Observation 9.3 — Zeta Regularisation as Extended Carrier**

1+2+3+…=∞ in the direct summation carrier. But the gradient field $G_\zeta$ (the function that equals $\Sigma\ 1/n^s$ where it converges) extends by analytic continuation to a larger carrier, giving $\zeta(-1)=-12$.

This is the same move as $\sqrt{(-1)}$: not defined in the smaller carrier, meaningful in the extended one. The result $\zeta(-1)=-12$ is not the sum of 1+2+3+…; it is the value of the analytically-continued gradient field at $s=\sqrt{1}$. Verified via functional equation: $2^{-1}\cdot\pi^{-2}\cdot\sin(-\pi/2)\cdot\Gamma(2)\cdot\zeta(2) = -1/12$ exactly.

## 10. The Mechanism in Action: Further Connections

Computational play sessions extended the mechanism across several domains. The repository contains complete verified code for all claims below.

### 10.1 Chaos: The Logistic Map as Observation 2.2 Made Visible

The logistic map $x \to r\cdot x\cdot(1-x)$. Fixed point $x^* = 1-1/r$ is stable for $r < 3$. At r=3: demand exceeds coherence threshold, pattern reconfigures to period-2 orbit. Each subsequent bifurcation is a further reconfiguration as demand grows. The Feigenbaum point r≈3.570 is Observation 2.2: infinite period doublings complete, no finite threshold contains the demand. The periodic windows in the chaotic regime (e.g., r≈3.83) are SIMPLER patterns propagating outward from the chaotic core — exactly Observation 2.2's second clause. Feigenbaum's constant $\delta \approx 4.6692$ is verified computationally from successive bifurcation intervals.

### 10.2 Mandelbrot Set as Coherence Boundary

The iteration $z \to z^2+c$ in $\mathbb{C}$ is propagation in the complex carrier. The Mandelbrot set $M = \{c \in \mathbb{C} : \text{orbit of 0 stays bounded}\}$ is the set of c for which propagation under $G_c$: $z \to z^2+c$ remains coherent. The boundary $\partial M$ is the locus of c at exactly the coherence threshold — infinite complexity because it traces the exact interface of Observation 2.2. The Mandelbrot set is a coherence map of the complex carrier indexed by gradient field $G_c$.

### 10.3 Central Limit Theorem as Coherence Attractor

The Gaussian N(0,1) is the maximum entropy distribution on $\mathbb{R}$ with fixed variance — the coherence state of the probability carrier (Play 3: Gaussian integral = $\sqrt{\pi}$). The CLT says: the Gaussian is an ATTRACTOR in the probability carrier under convolution. Under convolution, entropy increases (entropy power inequality). The maximum entropy state is Gaussian.

Therefore repeated averaging reconfigures every distribution toward the Gaussian. Verified computationally: uniform, bimodal (±2), exponential, and Bernoulli distributions all converge to Gaussian as n_avg increases, with skewness and excess kurtosis approaching 0. The CLT is the Second Law of Thermodynamics for probability distributions.

## 10.4 Catalan Numbers as Coherent Path Counts

$C_n = (1/(n+1))\cdot C(2n, n) = 1, 1, 2, 5, 14, 42, \ldots$ counts Dyck paths of semilength n: paths from (0,0) to (2n,0) with steps +1 (load +1) and −1 (reconfiguration −1), never going below 0 (coherence maintained throughout). $C_n$ IS the count of ways to compose n propagation steps while staying coherent. Recursive structure: every coherent path decomposes uniquely into first-return sub-paths — the propagation hierarchy in miniature. Asymptotically $C_n \sim 4^n/(n^{3/2}\sqrt{\pi})$ — the $\sqrt{\pi}$ from the Gaussian integral appears again ($G_{rot}$ coherence constant). Verified: Dyck path counts match formula for n = 0 through 8.

## 10.5 Noether's Theorem: Symmetry Gradients Generate Conservation Laws

A symmetry of the action $S = \int L\, dt$ is a gradient field $G_{sym}$ that leaves S invariant (S is a fixed point of $G_{sym}$). When S is a $G_{sym}$ fixed point, the quantity $J = \partial L/\partial \dot{q} \cdot \delta q_{sym}$ is conserved along every trajectory.

**Noether's Theorem as Fixed-Point Condition**

**$G_{time}$:** time translation leaves S invariant → Hamiltonian H (energy) conserved
**$G_{space}$:** spatial translation leaves S invariant → momentum p conserved
**$G_{rot}$:** rotation leaves S invariant → angular momentum L conserved
**U(1) gauge:** phase rotation leaves S invariant → electric charge conserved

Every conservation law IS a gradient family leaving the action invariant. 'Things that don't change' are patterns fixed by their gradient. Conservation laws are fixed-point conditions, not separate postulates. Verified: harmonic oscillator energy conserved to $2.5\times10^{-5}$ over 1000 steps (numerical integration error only).

## 10.6 Isoperimetric Inequality: The Circle as G

Among all closed curves of perimeter L, the circle encloses maximum area $A = L^2/(4\pi)$. Equality iff circle. In mechanism terms: the circle has constant curvature $\kappa = 1/r$ at every point — no point has more gradient demand than any other. It is the uniform-load enclosure, the maximum entropy shape for given perimeter. The factor $4\pi = (2\pi)^2/\pi$ is $G_{rot}$'s coherence cycle squared, again. Verified: isoperimetric ratio $L^2/(4\pi A)$ decreases toward 1 as polygon sides increase, reaching 1 for the circle.

## 10.7 The Collatz Conjecture: Probabilistic Coherence Without Proof

The Collatz map: C(n) = n/2 if n even, 3n+1 if odd. Verified for all $n < 2^{68}$. No proof exists.

**Mechanism Analysis of Collatz**

Average load change per step in the multiplicative carrier: $E[\Delta \ln(n)] = (1/2)(-\ln 2) + (1/2)(\ln 3 - \ln 2) = \ln(3/4)/2 \approx -0.144$. Negative drift → sequences shrink on average.

Measured average: −0.152 (verified computationally for n=3 to 1000, 100 steps each). Theoretical: −0.144. Both confirm negative drift.

Why this is not a proof: the variance is large; some sequences reach 342× their starting value before descending (n=27 reaches 9232). The gap between PROBABILISTIC convergence (negative average drift almost surely forces convergence) and DETERMINISTIC convergence (every specific sequence must converge) requires understanding exact number-theoretic structure of the multiplicative/additive carrier interaction. Until then, the mechanism can say 'almost certainly' but not 'necessarily'.

## 11. The Full Propagation Hierarchy

| Carrier V | Gradient Family Γ | System | What is forced |
|---|---|---|---|
| {0,1} | Full Boolean | Classical logic | LNC, LEM, Modus Ponens |
| {0,1} | Constructive | Intuitionistic | LEM unreachable |
| ℕ | Successor + add | Arithmetic | Peano axioms as step counts |
| ℤ | ℕ + reconfiguration | Integers | Additive inverses as signed steps |
| ℚ | ℤ + ratio (Stern-Brocot) | Rationals | Density; mediant = min-load fraction |
| ℝ | ℚ + coherence | Reals | Cauchy completion (Type B) |
| ℂ | ℝ + $G^2=G_{neg}$ | Complex | G_rot forced; sin/cos as $G_{diff}^4$ orbits |
| Prob. carrier | Convolution | Probability | Gaussian as coherence attractor (CLT) |
| Dual(ℝ) | ℝ + ε | Calculus 1st | Product/chain rule; FTC |
| n-Dual(ℝ) | ℝ + n directions | Taylor order n | 1/k! from commutativity |
| Partial-drag | ℝ + q-step | q-Calculus | [n]_q deformed counting |
| {0,1}+state | Linear G family | Linear logic | Consumption constraint |
| {0,1}+tags | Relevance G | Relevance logic | Gradient connectivity |
| Solvable group | Abelian chain | Radical solutions | Zero drag → solvable (deg ≤4) |
| $A_5$-carrier | Non-abelian | Quintic | Irreducible drag → unsolvable |
| TM states | Exec. history | Computation | Halting = fixed point; H(H) = Obs. 2.2 |
| Sequence carrier | Diagonal | Set theory | Uncountability = Obs. 2.2 |
| Mult. carrier | G_× (primes) | Number theory | FTA; PNT; ζ = partition function |
| Function space | Euler-Lagrange | Mechanics | Reconfiguration = conservation laws |

*Table 11.1: The propagation hierarchy. Each level is forced by demands of the previous.*

## 12. Computational Verification

The complete implementation is available in the repository. All gradient fields are functions applied to patterns, not methods on number objects. The following listings show the core implementation, stripped of exploratory print statements.

### 12.1 Core Primitives

```
from dataclasses import dataclass, field
from typing import FrozenSet, Any

@dataclass
class Pattern:
    val:          Any
    load:         float
    available:    bool            = True
    history_tags: FrozenSet[str]  = field(default_factory=frozenset)

    def consume(self) -> 'Pattern':
        if not self.available:
            raise LinearLogicViolation('Pattern already consumed')
        return Pattern(self.val, self.load, False, self.history_tags)

    def connected_to(self, other: 'Pattern') -> bool:
        if not self.history_tags or not other.history_tags: return False
        return bool(self.history_tags & other.history_tags)

@dataclass
class Context:
    threshold:   float = 1.0
    drag_regime: str   = 'zero'

    def combined_load(self, p: Pattern, q: Pattern) -> float:
        if self.drag_regime == 'zero':      return p.load + q.load
        elif self.drag_regime == 'linear':
            p.consume(); q.consume()  # raises if either consumed
            return p.load + q.load
        elif self.drag_regime == 'relevance':
            if not p.connected_to(q):
                raise RelevanceLogicViolation('No shared gradient path')
            return p.load + q.load
```

### 12.2 Gradient Fields

```
def G_neg(p):  # unary: preserves load
    return Pattern(1-p.val, p.load, p.available,
                   p.history_tags | frozenset(['neg']))

def G_and(p, q, ctx=None):
    ctx = ctx or C
    return Pattern(int(p.val)*int(q.val),
        ctx.combined_load(p, q),  # context determines drag regime
        history_tags=p.history_tags | q.history_tags | frozenset(['and']))
```

```
def G_or(p, q):
    return Pattern(max(int(p.val), int(q.val)),
        min(p.load, q.load),  # easiest path: minimum load
        history_tags=p.history_tags | q.history_tags | frozenset(['or']))
```

## 12.3 Dual Numbers, Taylor, and High-Order

```
import math

class Dual:
    def __init__(self, v, e=0.): self.v, self.e = float(v), float(e)
    def __mul__(self, o):
        # Leibniz: derived from co-propagation; e1*e2 = 0 boundary condition
        return Dual(self.v*o.v, self.e*o.v + self.v*o.e)
    def exp(self):
        ev = math.exp(self.v); return Dual(ev, self.e*ev)  # val ≡ e-component
    def sin(self): return Dual(math.sin(self.v),  self.e*math.cos(self.v))
    def cos(self): return Dual(math.cos(self.v), -self.e*math.sin(self.v))

class HighOrderDual:
    def exp(self):
        # CRITICAL: exp(x₀+u) = exp(x₀) × exp(u)
        # G_val and G_ε are SEPARATE gradient applications
        # Conflating them is the reification error (v and L must not be conflated)
        e0    = math.exp(self.coeffs[0])  # G_val: x₀ → exp(x₀)
        exp_u = self._nilpotent_exp()      # G_ε:  u → exp(u) as polynomial
        return HighOrderDual([e0 * exp_u[i] for i in range(self.n+1)])
```

## 12.4 Number Systems

```
class N:  # propagation step counts
    def __init__(self, steps: int): self.steps = steps
    def successor(self): return N(self.steps + 1)
    def __add__(self, o): return N(self.steps + o.steps)

class Q:  # load ratios
    def __init__(self, p, q=1):
        g = gcd(abs(p), abs(q)); sign = -1 if q<0 else 1
        self.p = sign*p//g; self.q = sign*q//g
    def midpoint(self, other): return (self + other) / Q(2)

# R as coherence completion: sqrt(2) via Newton reconfiguration in Q
def sqrt2_real():
    def gen(n):  # n-th Newton iterate: all in Q
        x = Q(1)
        for _ in range(n): x = (x + Q(2)/x) / Q(2)
        return x
    return CauchyReal(gen, 'sqrt2')

# C from G^2 = G_neg (Type D derivation)
def G_rot(a, b): return (-b, a)  # G_rot^2 = G_neg: verified
# G_rot(G_rot(3,1)) = (-3,-1) = G_neg(3,1)  =>  G_rot^2 = G_neg
```

## 12.5 Integration, FTC, and Solve-to-Coherence

```
def integrate(f, a, b, theta=1e-10):  # coherence accumulation, no limits
    n, prev = 4, None
    for _ in range(51):
        h = (b-a)/n
        total = sum(f(a+(i+0.5)*h) for i in range(n)) * h
        if prev is not None and abs(total-prev) < theta: return total
        prev, n = total, n*2
    return prev

def ftc_verify(f, x, a=0.):  # G_diff ∘ G_int = G_id
    eps = 1e-7
    return {
        'G_diff(G_int(f))': (integrate(f,a,x+eps) - integrate(f,a,x)) / eps,
        'f(x)': f(x),  # G_id applied directly
        'identity': 'G_diff ∘ G_int = G_id  (same structure as ¬¬P = P)',
    }

def solve_to_coherence(solver_step, input_val, tol=1e-12):
    # Gradient arrives WITH solution (Pugmire 2026b)
    # Memory: constant 48 bytes regardless of iteration depth
    class FP:  # flux pattern: (val, flux) co-propagating
        def __init__(self, v, f=0.): self.val, self.flux = float(v), float(f)
        def __add__(self, o):
            if not isinstance(o,FP): o=FP(o,0.)
            return FP(self.val+o.val, self.flux+o.flux)
        def __mul__(self, o):
            if not isinstance(o,FP): o=FP(o,0.)
            return FP(self.val*o.val, self.val*o.flux+o.val*self.flux)
        def __truediv__(self, o):
            if not isinstance(o,FP): o=FP(o,0.)
            return FP(self.val/o.val,
                      (self.flux*o.val-self.val*o.flux)/o.val**2)
        def __pow__(self, n): return FP(self.val**n, self.flux*n*self.val**(n-1))
    a = FP(input_val, 1.0)  # unit flux seeds d/da
    state = FP(1.0, 0.0)    # zero flux: no gradient history yet
    for i in range(500):
        prev = state
        state = solver_step(state, a)
        if abs(state.val - prev.val) < tol:
            return state.val, state.flux, i+1
    return state.val, state.flux, 500
```

## 12.6 Halting Problem and Play Session Highlights

```
def mandelbrot_coherent(c, max_iter=100, escape=2.0):
    # Points where z -> z^2+c stays bounded = coherent under G_c
    z = 0+0j
    for i in range(max_iter):
        z = z*z + c
        if abs(z) > escape: return False, i
    return True, max_iter

# H(H): self-referential load doubling
L = 1.0
for d in range(8):  # L_d = 2^d -> infinity
    L = L + L  # H_H includes H_H: each depth requires full prior history
```

```
def collatz_stopping_time(n):  # Observation: negative average drift
    count = 0
    while n != 1:
        n = n//2 if n%2==0 else 3*n+1
        count += 1
    return count

def golden_reconfigure(x0=1.0, steps=25):
    x = x0  # phi = fixed point of x -> 1 + 1/x
    for _ in range(steps): x = 1 + 1/x
    return x  # converges to (1+sqrt(5))/2 = 1.618...

def logistic_attractor(r, n_discard=500, n_iter=200):
    x = 0.5  # period doublings at r=3, 3.449, 3.544, 3.570...
    for _ in range(n_discard): x = r*x*(1-x)
    return [x := r*x*(1-x) for _ in range(n_iter)]

def catalan(n):  # count coherent propagation paths of semilength n
    return math.comb(2*n, n) // (n+1)

def euler_product_sin(x, n_terms=1000):
    # sin(pi*x) = pi*x * product(1 - x^2/n^2)
    # Zeros at integers are seeds of G_rot; sum 1/n^2 = pi^2/6
    result = math.pi * x
    for n in range(1, n_terms+1):
        result *= (1 - (x/n)**2)
    return result
```

## 13. Conclusion

This paper derived the natural numbers, integers, rationals, reals, complex numbers, dual numbers, the differential and integral operators, the Fundamental Theorem of Calculus, Taylor coefficients, drag regimes, flux propagation, and the structural account of the Halting Problem from a single mechanism: differential propagation of relational patternings through gradient fields accumulating loaded history.

Extended play with the mechanism revealed further connections. sin and cos are the real projections of the rotation gradient's 4-step orbit in $\mathbb{C}$ — not primitive functions but $G_{diff}^4$ fixed points. The three constants e, π, φ are structurally distinct: $G_{diff}$ fixed point, $G_{rot}$ coherence cycle, and ratio reconfiguration fixed point respectively — three different types, not three instances of the same mystical harmony. The Gaussian integral $\int e^{-x^2}dx = \sqrt{\pi}$ is their collision at the quadratic carrier. Primes are seeds of the multiplicative carrier; FTA is unique decomposition of loaded history into seeds; PNT is the statement that seeds thin as propagation cost grows. The Mandelbrot set is the coherence boundary of iterated complex propagation. The CLT is gradient descent toward the maximum entropy state (Gaussian) under convolution. Noether's theorem states that conservation laws are gradient families leaving the action invariant. The unsolvability of the quintic follows from $A_5$ generating irreducible non-commutative drag.

The key results, with claim types:

**(A) Derives:** $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, linear drag consumption state, relevance drag connectivity, flux propagation.

**(B) Motivates and constructs:** $\mathbb{R}$ as coherence completion; drag regimes as boundary conditions.

**(C) Reveals structure:** FTC as $G_{diff} \circ G_{int} = G_{id}$; Fourier/Taylor as the same expansion in different gradient families; Noether as gradient invariance; Collatz as probabilistic coherence without deterministic proof.

**(D) Asks the right question:** $\mathbb{C}$ from $G^2=G_{neg}$; p-adics from prime-centric drag; zeta regularisation as carrier extension.

**Mathematical Truth as Coherence**

A mathematical theorem is a fixed point of the proof process. If T is true, any valid gradient field in the proof carrier leaves T stable. The loaded history of T cannot be reduced by further scrutiny.

This is not a definition of truth. It is an observation about what the proof process does. The constants are the least unstable structures — patterns that propagation cannot eliminate. The theorems are boundary conditions. The impossibility results (Halting, Gödel, quintic) are coherence failures.

The ‘unreasonable effectiveness of mathematics in describing nature’ (Wigner) is less mysterious through this lens: nature uses gradient fields to propagate patterns, and mathematics studies gradient fields and their fixed points. The effectiveness is not unreasonable — mathematics and physics study the same structures in different carriers. P/G → Q generates both.

Mathematics is the science of what cannot be changed. A theorem is a pattern that propagation cannot destroy. Truth is coherence.

Three open problems through the mechanism:

**Riemann Hypothesis:** zeros of ζ are where multiplicative and additive loaded histories cancel in $\mathbb{C}$. The functional equation forces reflection symmetry about Re(s)=1/2. The hypothesis says all non-trivial zeros lie on this line — that the coherence structure of ζ is palindromic about the critical line.

**P vs NP:** verification = checking if P reaches coherence (polynomial); solution = finding the path to coherence. P≠NP (expected) would mean finding the path generally requires more loaded history than checking it. Reconfiguration is harder than coherence testing.

**Collatz:** negative average drift is verified but insufficient for proof. The gap between probabilistic and deterministic convergence requires understanding exact number-theoretic correlations between the multiplicative and additive carriers.

Euler’s identity as coda: $e^{i\pi} + 1 = 0$. Three different kinds of fixed points (e: $G_{diff}$ fixed point; i: $G^2=G_{neg}$ carrier extension; π: $G_{rot}$ coherence cycle), co-propagated in $\mathbb{C}$, meeting at the additive identity. Beautiful because forced, not chosen. The mechanism at its most compressed.

**Mathematical Truth as Carrier-Relative Coherence**

A theorem is a pattern coherent in its carrier. Truth is carrier-relative coherence.

The Gödel sentence G is not coherent in C (not provable within C). But G is coherent in the

meta-carrier that can reason about C. This is not a contradiction; it is the mechanism generating the next carrier in the hierarchy — the same pressure that generates $\mathbb{R}$ from $\mathbb{Q}$, $\mathbb{C}$ from $\mathbb{R}$.

Incompleteness is the condition where coherence requires a larger carrier than the system provides. √2 is not in $\mathbb{Q}$ but is in $\mathbb{R}$; G is not provable in C but is provable from outside C. Both are carrier extension under coherence pressure. The incompleteness theorems do not defeat the framework — they are the framework applied to formal systems: every sufficiently expressive system generates patterns whose coherence demands a larger carrier.

Mathematics is the science of what cannot be changed relative to a carrier. A theorem is a pattern that propagation cannot destroy within that carrier. Truth is coherence. Incompleteness is the boundary condition where one carrier ends and a larger one begins.

And the gradient fields through which patterns propagate — $G_{diff}$, $G_{rot}$, $G_{\times}$, the logical connectives, the symmetry families of Noether's theorem — are themselves patterns in a meta-carrier. The distinction between pattern and gradient is local and momentary. The mechanism that generates all of mathematics is itself an instance of the mechanism. P/G → Q generates the description of P/G → Q. There is no outside.

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*Complete implementation and play session code: github.com/[repository]*